INTE 22303 : Artificial Intelligence (24/25)

Assignment 1

Duration: 90 minutes

Overview

In this assignment, you will build a lecture hall booking workflow using n8n.

You will start from a starter n8n workflow JSON that contains only the booking form. You must build the remaining workflow nodes.

In the end you need to submit

1. **Export Json file.**
2. **Screenshot of the complete n8n workflow**
3. **Screenshot of the booking form**
4. **Screenshot of a created Google Calendar event**
5. **Screenshot of the confirmation email**
6. **Available booking evidence**
7. **Unavailable booking evidence**

Scenario

A requester submits a form to book Lecture Hall A. The workflow must check whether the lecture hall is available for the selected date and time.

If the hall is available, the workflow should:

1. Create a Google Calendar event.
2. Store the booking details inside the Calendar event.
3. Use Gemini to generate a short confirmation email.
4. Send the confirmation email using Gmail.
5. Show a success page to the requester.

If the hall is unavailable, the workflow should:

1. Show an unavailable message to the requester.

Starter Workflow

You will receive a starter n8n JSON file containing only the form trigger.

The form will contain these fields:

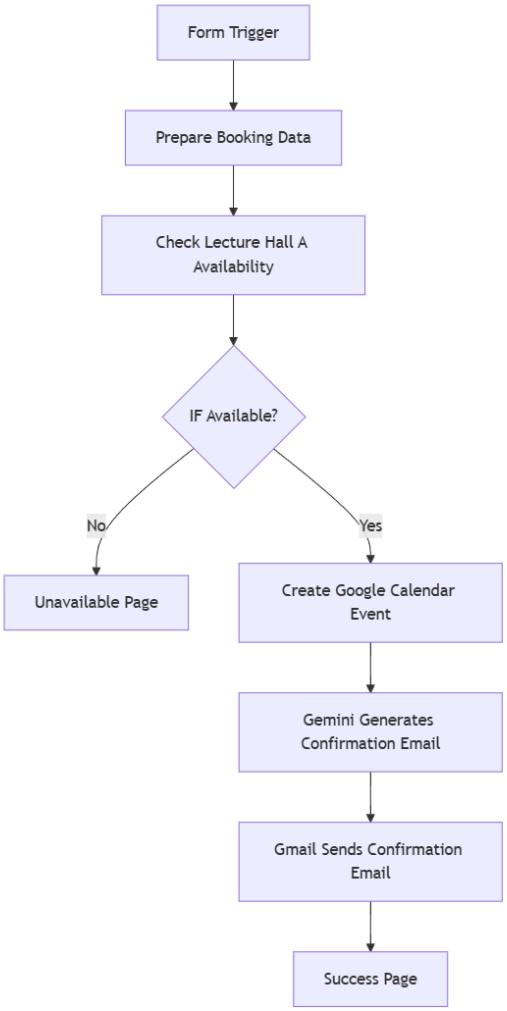
Field	Type	Field Name
Requester Name	Text	requesterName
Requester Email	Email	requesterEmail
Booking Date	Date	bookingDate
Start Time	Dropdown	startTime
End Time	Dropdown	endTime

The start time options are: from 8.00 am to 17.00 pm by hour.

The end time options are: from 9.00 am to 18.00 pm by hour.

Required Workflow

Your final workflow should follow this structure:



Required Nodes

Your workflow must use the following nodes:

No.	Node
1	Edit Fields
2	Google Calendar
3	IF
4	Google Calendar
5	Basic LLM Chain
6	Google Gemini Chat Model
7	Gmail
8	Form Ending

Assignment Questions

Answer the following questions by building the required parts of the n8n workflow.

The required n8n expressions are provided under the relevant questions.

In the expressions below, replace placeholder node names with the exact names used in your own workflow:

Question 1: Booking Data Preparation

Build the part of the workflow that prepares the form data for Calendar use.

Your workflow must create:

New Field	Required Value
startDateTime	ISO date-time value created from bookingDate and startTime
endDateTime	ISO date-time value created from bookingDate and endTime

Use the timezone:

```
Asia/Colombo
```

Use these expressions:

startDateTime

```
{{ DateTime.fromISO($json.bookingDate + 'T' + $json.startTime, { zone: 'Asia/Colombo' }).toISO() }}
```

endDateTime

```
{{ DateTime.fromISO($json.bookingDate + 'T' + $json.endTime, { zone: 'Asia/Colombo' }).toISO() }}
```

Question 2: Calendar Availability Check

Build the part of the workflow that checks whether Lecture Hall A is free for the requested period.

Your workflow must:

- Use the Lecture Hall A Google Calendar.
- Check availability between startDateTime and endDateTime.
- Use the Asia/Colombo timezone.
- Return an availability result that can be used by the next part of the workflow.

Use these values:

Time Min

```
{{ $json.startDateTime }}
```

Time Max

```
{{ $json.endDateTime }}
```

Question 3: Availability Decision

Build the decision point of the workflow.

Question 4: Calendar Booking Event

Build the part of the workflow that creates the confirmed booking in Google Calendar.

Your Calendar event must include:

Event Field	**Required Content**
Calendar	Lecture Hall A
Start	Prepared start date-time
End	Prepared end date-time
Summary	Lecture hall booking with requester name
Description	Booking details listed below
Location	Lecture Hall A
Availability status	Busy

The Calendar event description must include:

- Requester name
- Requester email
- Lecture hall name
- Booking date

- Start time
- End time
- Booking status
- Submission time, if available

Use these expressions:

Start

```
{{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.startDateTime }}
```

End

```
{{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.endDateTime }}
```

Summary

```
{{ 'Lecture Hall Booking - ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.requesterName }}
```

Description

```
{{
'Requester Name: ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.requesterName + '\n' +
'Requester Email: ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.requesterEmail + '\n' +
'Lecture Hall: Lecture Hall A\n' +
'Booking Date: ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.bookingDate + '\n' +
'Start Time: ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.startTime + '\n' +
'End Time: ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.endTime + '\n' +
'Status: Confirmed\n' +
'Submitted At: ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.submittedAt
}}
```

Question 5: AI Confirmation Email Content

Build the part of the workflow that uses Gemini to generate a confirmation email body.

The generated email must:

- Be short and professional.
- Use only the submitted booking information.
- Mention Lecture Hall A.
- Include the booking date, start time, and end time.
- Not include invented institution names, policies, phone numbers, or extra details.
- Not include an email subject.
- End with this sentence:

```
This is an automatically generated confirmation from the lecture hall booking system.
```

Gemini must only generate the email message. It must not decide whether the hall is available.

Use this prompt:

```
Write a short professional lecture hall booking confirmation email.

Use only the supplied booking information.
Do not invent institution names, policies, phone numbers, or extra details.
Do not include an email subject.
Return only the plain-text email body.

Requester: {{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.requesterName }}
Lecture hall: Lecture Hall A
Date: {{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.bookingDate }}
Start time: {{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.startTime }}
End time: {{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.endTime }}

End with:
This is an automatically generated confirmation from the lecture hall booking system.
```

Question 6: Email Sending

Build the part of the workflow that sends the confirmation email.

Your workflow must:

- Send the email to the requester email address from the form.
- Use a suitable subject for a confirmed lecture hall booking.
- Use the Gemini-generated text as the email body.
- Send an email only when the booking is confirmed.

Use these expressions:

Send To

```
{{ $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.requesterEmail }}
```

Message

```
{{ $('GENERATE_AI_CONFIRMATION_NODE_NAME').item.json.text }}
```

Question 7: Success Response

Build the success response shown to the requester after a confirmed booking.

The success response must clearly state that:

- The booking was successful.
- The Calendar event was created.
- A confirmation email was sent.

You may include the requester name, date, start time, and end time in the response.

Optional message expression:

```
{{
'Your lecture hall booking was completed successfully.<br><br>' +
'<b>Requester:</b> ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.requesterName + '<br>' +
'<b>Date:</b> ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.bookingDate + '<br>' +
```

```
'<b>Time:</b> ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.startTime + ' - ' +
$('PREPARE_BOOKING_DATA_NODE_NAME').item.json.endTime + '<br><br>' +
'The booking has been added to Google Calendar and a confirmation email has been sent.'
}}
```

Question 8: Unavailable Response

Build the unavailable response shown to the requester when Lecture Hall A is already booked.

The unavailable response must clearly state that:

- Lecture Hall A is unavailable for the selected period.
- The requester should choose a different time.

This response must not create a Calendar event or send an email.

Optional message expression:

```
{{
'Lecture Hall A is already booked during the selected period.<br><br>' +
'<b>Date:</b> ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.bookingDate + '<br>' +
'<b>Time:</b> ' + $('PREPARE_BOOKING_DATA_NODE_NAME').item.json.startTime + ' - ' +
$('PREPARE_BOOKING_DATA_NODE_NAME').item.json.endTime + '<br><br>' +
'Please submit another request using a different time.'
}}
```

Bonus Mark Question: Time Validation

Build an optional validation part that checks whether the selected end time is later than the selected start time. if failed then show an invalid-time response when the selected time range is invalid.